

Translated ordered states and native Heisenberg collisions

Abstract

Differentiating the ordered multiplication kernel determines the products of translated states. These products define a flat differential module and retain the original ordered bar as its untranslated subcomplex. At a fixed nonzero Heisenberg level, weighted edge integrals give binary and ternary graded comparisons with the native chiral complex. Their defects determine minimal added collision modules for these prescribed maps. The ternary addition contains three pair-diagonal modules and a current module on the small diagonal. Exact sequences describe its cohomology relative to the original ordered source. A four-current calculation determines the next differential, supported on the three double-pair diagonals.

TRANSLATED ORDERED STATES AND NATIVE HEISENBERG COLLISIONS

1.1 THE TRANSLATION OF AN ORDERED PRODUCT

Fix $k \in \mathbb{C}^\times$ and $X = \mathbb{A}_{\mathbb{C}}^1$. All position functions are algebraic, and every differential operator has finite order. No completion or additional state differential is used. For two positions put $t = z_1 - z_2$ and $y = z_2$. The coefficient algebra is

$$\begin{aligned} C_2^o &= \{f \in \mathbb{C}[y, t, t^{-1}][u] : f(o), f(1) \in \mathbb{C}[y, t]\}, \\ C_2^1 &= \mathbb{C}[y, t, t^{-1}][u]du, \quad d_C f = \partial_u f du. \end{aligned}$$

The zero-jet ordered product is $a_1 a_2 = \omega_{12} b_{12}$, where $\omega_{12} = du/t$, $|a_i| = o$, and $|b_{12}| = -1$. Translations of the inputs must also differentiate ω_{12} . For example, a translation derivation L_1 with $L_1 z_1 = 1$ satisfies

$$(L_1 a_1) a_2 = -\frac{du}{t^2} b_{12} + \frac{du}{t} L_1 b_{12}. \quad (1.1)$$

The first summand is required even when b_{12} has a prescribed translation.

For $n = 2, 3$, write $R_n = \mathbb{C}[z_1, \dots, z_n]$. Here is a common definition of the coefficient algebra C_n . Let O_n be the total orders of the labels. For nonempty $A \subset O_n$, let R_A invert $z_i - z_j$ when the relative order of i, j varies in A . Let

$$\Omega_A = \mathbb{C}[u_\sigma, du_\sigma \mid \sigma \in A] / (\sum_A u_\sigma - 1, \sum_A du_\sigma).$$

The odd generators du_σ form an exterior algebra. A member of C_n is a family in $R_A \otimes \Omega_A$ compatible with every face restriction and coefficient localization. The differential sends u_σ to du_σ and fixes positions. Set

$$q_{ij} = \sum_{i \prec_\sigma j} u_\sigma, \quad \omega_{ij} = \frac{dq_{ij}}{z_i - z_j}.$$

On a face where the denominator is unavailable, $dq_{ij} = o$ and that component is zero. The position derivations $\partial_i = \partial_{z_i}$ preserve C_n and commute with d_C . Indeed, they preserve every localization and every face condition.

Adjoin closed symbols $a_i^{(r)}$ and $b_{ij}^{(r,s)}$, for $r, s \geq o$, of degrees o and -1 , respectively. Coefficients act graded centrally. Define the only nonzero nonunit products by

$$a_i^{(r)} a_j^{(s)} = \sum_{p=o}^r \sum_{q=o}^s \binom{r}{p} \binom{s}{q} (\partial_i^p \partial_j^q \omega_{ij}) b_{ij}^{(r-p, s-q)}, \quad i \neq j. \quad (1.2)$$

Products involving a b and another nonunit symbol are zero. Products of two a_i with the same label are zero. Let L_i act by ∂_i on coefficients. It increases the i -label jet index on a symbol and kills symbols without label i . The operators L_i commute. Total translation of a two-label symbol is $L_i + L_j$.

PROPOSITION 1.1.1. Formula (1.2) defines an associative differential graded algebra over C_n . Each L_i is a degree-zero derivation commuting with the differential. The untranslated symbols have exactly the product $a_i^{(o)} a_j^{(o)} = \omega_{ij} b_{ij}^{(o,o)}$.

Proof. Every product of three nonunit symbols vanishes under either parenthesization. This proves associativity. The coefficients in (1.2) are closed, since $d_C \omega_{ij} = 0$ and d_C commutes with position derivatives. Thus the differential obeys the graded Leibniz identity. Applying L_i differentiates the coefficient or increases the first b index. Pascal's identity combines these terms into (1.2) with r replaced by $r + 1$. The same calculation with $s + 1$ proves the L_j identity. Every other L_l fixes the kernel and both input labels. This proves the derivation assertion on all products. $\square$

1.2 THE RIGHT DIFFERENTIAL MODULE OF ORDERED STATES

Suspension lowers degree by one and satisfies $s(ca) = (-1)^{|c|} c sa$. The bar convention is $b_2(sa \otimes sb) = (-1)^{|a|} s(ab)$. For $n = 2$, the untranslated sector has symbols $e = [a_1|a_2]$ and $h = [b_{12}]$, both of degree -2 . Its differential is $be = -\omega_{12}h$ and $bh = 0$. For $n = 3$ and $\sigma = ijk$, use

$$e_\sigma = [a_i|a_j|a_k], \quad m_\sigma = [b_{ij}|a_k], \quad n_\sigma = [a_i|b_{jk}].$$

These eighteen symbols have degree -3 . Their free C_3 -module O_3 has differential

$$be_\sigma = -\omega_{ij}m_\sigma - \omega_{jk}n_\sigma, \quad bm_\sigma = bn_\sigma = 0, \quad b(cv) = d_C c v + (-1)^{|c|} c bv. \quad (1.3)$$

Denote the binary complex by O_2 .

Let P_n be the same bar sector with all translated symbols from (1.2). For a multi-index $\alpha \in \mathbb{N}^n$, write v_α for the symbol whose input-label jets are α . Each P_n is the direct sum of $C_n v_\alpha$ over its untranslated symbols and finite jet indices. On these words,

$$L_i(cv_\alpha) = (\partial_i c)v_\alpha + cv_{\alpha+e_i}.$$

For the three-letter symbol the bar differential is explicitly

$$bv_\alpha = L^\alpha bv_0, \quad b(e_\sigma)_\alpha = - \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} \left((\partial^\gamma \omega_{ij})(m_\sigma)_{\alpha-\gamma} + (\partial^\gamma \omega_{jk})(n_\sigma)_{\alpha-\gamma} \right). \quad (1.4)$$

Mixed words have zero symbol differential. The binary formula omits the second summand. The graded coefficient rule in (1.3) remains in force.

Let D_n be the Weyl algebra of R_n , with $[\partial_i, z_j] = \delta_{ij}$. The right action on P_n is

$$w\partial_i = -L_i w, \quad wf = fw \quad (f \in R_n).$$

The identity $(w\partial_i)f - (wf)\partial_i = w(\partial_i f)$ verifies the Weyl relation. Flatness of the L_i verifies all derivative commutators. Equation (1.4) gives $bL_i = L_i b$, so P_n is a complex of right D_n -modules. The untranslated inclusion $O_n \hookrightarrow P_n$ is a cochain map over R_n .

PROPOSITION 1.2.1. There is an isomorphism of right differential-module complexes

$$O_n \otimes_{R_n} D_n \longrightarrow P_n, \quad x \otimes A \longmapsto xA. \quad (1.5)$$

Consequently,

$$H^r(P_n) = H^r(O_n) \otimes_{R_n} D_n. \quad (1.6)$$

This is not an identification of P_n with O_n .

Proof. Balancing follows from the stated regular-function action. The ordered monomials ∂^α form a free left R_n -basis of D_n . For $x = cv_0$, the image of $x \otimes \partial^\alpha$ is $(-1)^{|\alpha|} cv_\alpha$ plus terms with smaller jet order. This triangular formula gives an inverse on every finite sum. Right linearity is built into the definition. The chain identity follows from $bL_i = L_ib$. Finally, left R_n -freeness of D_n makes tensoring exact, proving (1.6). $\square$

The products (1.2) specify what the induction means for translated states. Keeping the same formula $a_i^{(r)} a_j^{(s)} = \omega_{ij} b_{ij}^{(r,s)}$ for every jet would contradict (1.1).

1.3 THE UNEXTENDED HEISENBERG TARGET

The Heisenberg states, vacuum, current, and translation are

$$V_k = \mathbb{C}[x_1, x_2, \dots], \quad \mathbf{I} = \mathbf{I}, \quad J = x_1, \quad T = \sum_{r \geq 1} r x_{r+1} \partial_{x_r}.$$

Every state is a finite polynomial, is even, and has degree zero. The modes are $J_{-r} = x_r$, $J_r = kr \partial_{x_r}$ for $r \geq 1$, and $J_0 = 0$. The field of a monomial is the normally ordered product of the divided current derivatives. Normal ordering puts multiplication operators before derivative operators. Commuting these operators gives the finite Wick pairing rule. For factors x_r and x_l , a contraction is

$$C_{rl}(t) = \frac{k(-1)^{r-1}(r+l-1)!}{(r-1)!(l-1)!} t^{-r-l}.$$

Sum over all partial pairings of distinct input factors. Each pair contributes its contraction, and the unpaired factors remain normally ordered. The rule follows by applying $AB = BA + [A, B]$ until every multiplication operator precedes every derivative operator. Each contracted factor is removed, so every fixed monomial has finitely many pairings. In particular,

$$p(t) = Y(J, t)J = \frac{k}{t^2} \mathbf{I} + (e^{tT} J)J, \quad Y(\mathbf{I}, t)a = a, \quad T^r J = r! x_{r+1}.$$

The right differential module $\mathcal{H}_k = \omega_X \otimes V_k$ has action $(f a dz) \partial_z = -(f' a + f T a) dz$. For $t = z - w$ and $y = w$, its chiral operation is

$$\mu(f(z, w)(a dz \boxtimes b dw)) = \text{pp}_t(f(y + t, y)Y(a, t)b), \quad \text{pp}_t \frac{a(y)}{t^{r+1}} = \frac{a(y)dy}{r!} \delta \partial_t^r. \quad (1.7)$$

The orientation is $dz \wedge dw = dt \wedge dy$. Multiplication by a Laurent coefficient occurs before taking the principal part. An existing principal-part class only admits multiplication by regular functions. The normal action is $-\partial_t$ on Laurent representatives. In particular, $(\partial \partial_t^r)t = r \partial \partial_t^{r-1}$.

Translation of fields follows directly from the mode formulas: $Y(Ta, t) = \partial_t Y(a, t)$ and $TY(a, t)b = \partial_t Y(a, t)b + Y(a, t)Tb$. These two equations make (1.7) a right differential-module map. The contraction relation $C_{lr}(-t) = C_{rl}(t)$ gives vertex skew symmetry by interchanging the two inputs in the Wick sum. Normal coordinate transposition acts as $p(t, y) \mapsto -\text{pp}_t(e^{tT} p(-t, y + t))$. The density minus sign therefore gives skew symmetry for the chiral operation.

For three monomial states the Wick expression belongs to $V_k[[u, v]][u^{-1}, v^{-1}, (u - v)^{-1}]$. Its two radial expansions are the two ordered products. Pairing the first two inputs first gives its expansion along $u = v$, hence the iterate. The following residue identity gives their difference:

$$\begin{aligned} \text{Res}_v \text{Res}_t G(v + t, v) dt dv &= \text{Res}_u \text{Res}_v i_{u,v} G(u, v) dv du \\ &\quad - \text{Res}_v \text{Res}_u i_{v,u} G(u, v) du dv. \end{aligned}$$

Here $i_{u,v}$ expands in nonnegative powers of v/u . For a rational numerator the identity is the partial-fraction identity for the poles $u = 0$ and $u = v$. For a power-series numerator, each residue uses finitely many coefficients, reducing to that case. Apply the identity after multiplication by every polynomial testing the two normal variables. This proves chiral Jacobi coefficient by coefficient, including input translations and finite coefficient poles.

For a partition π of $\{1, \dots, n\}$, let Δ_π identify its blocks and let j_π exclude collisions between distinct blocks. The native complex is

$$E_n = \bigoplus_{\pi} (\Delta_\pi)_* (j_\pi)_* j_\pi^* (\boxtimes_{B \in \pi} \mathcal{H}_k[1]). \quad (1.8)$$

The closed direct image uses the right transfer module with all finite normal derivatives. There is no additional degree shift. Formula (1.8) is the Chevalley–Cousin carrier of [1], §3.4.II, equation (3.4.II.1). Its partition summand has degree $-\#\pi$. For even states the collision differential is

$$Q[a|b] = [\mu(a, b)], \quad Q[a|b|c] = [\mu_{12}(a, b)|c] - [\mu_{13}(a, c)|b] + [\mu_{23}(b, c)|a]. \quad (1.9)$$

Every map includes transverse localization and the density permutation. Swapping suspended even blocks changes the sign. The preceding Jacobi identity gives $Q^2 = 0$.

1.4 THE BINARY COMPARISON AND ITS SOURCE TERM

For an actual edge coefficient define

$$\begin{aligned} A(g) &= \int_0^1 g(u) du, & M(g) &= \int_0^1 u g(u) du, \\ q_g &= \text{pp}_t(t p(t) A(g)), & v_g &= \text{pp}_t(p(t) M(g)). \end{aligned} \quad (1.10)$$

Characteristic zero makes these polynomial integrals exact. Every normal coefficient is regular in the remaining block positions. Integration by parts gives

$$v_{\partial_u f} + q_{f/t} = \text{pp}_t(p(t) f(1)), \quad q_{\partial_u f} = k \text{pp}_t \frac{f(1) - f(0)}{t} \mathbf{1}. \quad (1.11)$$

The second identity uses endpoint regularity and the factor t in q_g .

Define a degree-zero R_2 -linear map $B_2 : O_2 \rightarrow E_2$ by

$$B_2(fe) = f(1)[J_1|J_2], \quad B_2(gdu e) = [v_g], \quad B_2(fh) = 0, \quad B_2(gdu h) = -[q_g].$$

Let N_2 be the right differential module generated by $\partial_{12} \mathbf{1}$ in degree -1 . It is closed under Q , and

$$N_2 \cong D_2 / (t D_2 + \partial_y D_2). \quad (1.12)$$

Its independent forms are $f(y) \mathbf{1} \partial_{12} \partial_t^r$, $r \geq 0$.

Extend B_2 to P_2 by right differential linearity through (1.5). This is an actual map on translated states. For any base coefficient c and word v , its recursion is

$$B_n(cv_{\alpha+e_i}) = -B_n(cv_\alpha) \partial_i - B_n((\partial_i c) v_\alpha). \quad (1.13)$$

For an unmerged zero-form it sends the i -th jet to T on precisely the i -th native current. The identity holds for every jet order, because evaluation commutes with position differentiation.

PROPOSITION 1.4.1. The binary defect $\beta_2 = QB_2 - B_2b$ takes values in N_2 . On untranslated inputs its only nonzero component is

$$\beta_2(fh) = k \text{ pp}_t \frac{f(\mathbf{i}) - f(\mathbf{o})}{t} \mathbf{I}.$$

The complex

$$\tilde{P}_2 = P_2 \oplus N_2, \quad \tilde{b}(x, n) = (bx, \beta_2 x)$$

admits the native cochain map $F_2(x, n) = B_2x + n$. Among added submodules of the target for this fixed B_2 and splitting, N_2 is minimal.

Proof. For fe , the first identity in (1.11) gives zero defect. The target differential of a one-form image is zero by degree. For fh , the defect is $q_{\partial_u}f$, giving the displayed formula. Right differential linearity covers all jets. Since $Q^2 = b^2 = \mathbf{o}$, one has $Q\beta_2 + \beta_2b = \mathbf{o}$. The target differential is zero on N_2 , hence $\beta_2b = \mathbf{o}$. This proves $\tilde{b}^2 = \mathbf{o}$ and $QF_2 = F_2\tilde{b}$. Taking $f = u$ gives $k\delta_{12}\mathbf{I}$. Its right differential closure is all of (1.12), proving minimality. $\square$

The binary cohomology change is explicit in its lowest degree. Endpoint integration gives

$$H^{-2}(O_2) = R_2z \oplus R_2h, \quad z = te + uh.$$

Indeed, a cycle $fe + gh$ has $f \in R_2$ and $\partial_u g = f/t$. Regular endpoints force $t \mid f$, and subtracting a multiple of z leaves a constant multiple of h . Its connecting map sends z to $k\delta_{12}\mathbf{I}$ and h to zero. Therefore

$$H^{-2}(\tilde{P}_2) = z(tD_2 + \partial_y D_2) \oplus hD_2 \subset H^{-2}(O_2) \otimes_{R_2} D_2.$$

All other cohomology groups of the projection $\tilde{P}_2 \rightarrow P_2$ are isomorphic. Surjectivity of the displayed connecting map proves this assertion from the long exact sequence.

1.5 ALL THREE COLLISION DIAGONALS

For each word ijk , restrict coefficients to its edges $\{ijk, jik\}$ and $\{ijk, ikj\}$. Their coordinates are $u_L = q_{ij}$ and $u_R = q_{jk}$. Both coordinates equal one at the word vertex. For $c \in C_3^1$, write $c|_L = g_L du_L$ and $c|_R = g_R du_R$. Use (1.10) in each edge normal coordinate. Define $B_3 : O_3 \rightarrow E_3$ by

$$\begin{aligned} B_3(fe_{ijk}) &= f(ijk)[J_i|J_j|J_k], \\ B_3(ce_{ijk}) &= [v_{g_L}|J_k] - [J_i|v_{g_R}], \\ B_3(fm_\sigma) &= B_3(fn_\sigma) = \mathbf{o}, \\ B_3(cm_\sigma) &= -[q_{g_L}|J_k], \\ B_3(cn_\sigma) &= +[J_i|q_{g_R}]. \end{aligned} \tag{1.14}$$

Here f has degree zero and c has degree one. Every component of coefficient degree at least two has zero image. The mixed formulas are the binary images with the suspended spectator and its tensor sign. Extend to P_3 by (1.13).

For a finite inner principal part w , define $R_L(w) = Q[w|J]$ and $R_R(w) = Q[J|w]$. These operations retain every inner normal derivative. They take the outer principal part only after transverse localization. Put $\Delta_L f = f(ijk) - f(jik)$ and $\Delta_R f = f(ijk) - f(ikj)$.

LEMMA 1.5.1. The defect $\beta_3 = QB_3 - B_3b$ on untranslated words is

$$\begin{aligned}
\beta_3(fe_{ijk}) &= -[\mu_{ik}(f(ijk)(J_i, J_k))|J_j], \\
\beta_3(ce_{ijk}) &= R_L(v_{g_L}) - R_R(v_{g_R}), \\
\beta_3(fm_\sigma) &= k[\text{pp}_{t_{ij}}(\Delta_L f / t_{ij})\mathbf{1}|J_k], \\
\beta_3(fn_\sigma) &= -k[J_i|\text{pp}_{t_{jk}}(\Delta_R f / t_{jk})\mathbf{1}], \\
\beta_3(cm_\sigma) &= -R_L(q_{g_L}), \\
\beta_3(cn_\sigma) &= +R_R(q_{g_R}).
\end{aligned} \tag{1.15}$$

It vanishes on all higher coefficient degrees.

Proof. For a zero-form unmerged word, the two adjacent components cancel by the first identity in (1.11). The nonadjacent component has the middle unshuffle sign in (1.9). For a one-form unmerged word, all terms in B_3b restrict to two-forms on an edge and vanish. Its two outer collisions give the second line. On mixed zero-forms use the second identity in (1.11). On mixed one-forms the source differential again has zero edge image. The target differential gives the last two lines with the signs in (1.14). Higher coefficients vanish for the same edge-dimension reason. $\square$

1.6 THE MINIMAL TERNARY ADDED MODULE

Let $U_J = \text{span}_{\mathbb{C}}\{T^r J : r \geq 0\}$ with its stated translation. For each unordered pair ij with spectator l , let M_{ij} be the regular-coefficient right differential closure of $[\mathbf{1}\delta_{ij}|J_l]$ inside the $ij|l$ target summand. The pair block is listed first, and $t_{ij} = z_i - z_j$ uses $i < j$. Let S_3 be the right differential closure of $[J\delta_{123}]$ in the small-diagonal summand. Then

$$\begin{aligned}
M_{ij} &\cong D_3 / (t_{ij}D_3 + (\partial_i + \partial_j)D_3), \\
S_3 &\cong D_3 / (t_{12}D_3 + t_{23}D_3).
\end{aligned}$$

The vacuum fixes the first tangent relation. The spectator derivatives generate the independent states $T^r J$. For S_3 , the total tangent derivative generates those same independent states, so there is no tangent annihilator. Normal ordering in the Weyl algebra gives independent finite transfer forms in both presentations. This proves completeness of their relations.

Put $M_3 = M_{12} \oplus M_{13} \oplus M_{23}$ in degree -2 and put S_3 in degree -1 . Their direct sum $N_3 = M_3 \oplus S_3$ has zero differential. Indeed, a pair term has vacuum in its merged block and regular outer coefficients. Its remaining principal part is zero by $Y(\mathbf{1}, s)J = J$. Right differential linearity preserves this vanishing. The small-diagonal term has no further collision.

THEOREM 1.6.1. The defect β_3 has values in N_3 . The right differential-module complex

$$\bar{P}_3 = P_3 \oplus N_3, \quad \bar{b}(x, n) = (bx, \beta_3 x) \tag{1.16}$$

admits the native cochain map $F_3(x, n) = B_3 x + n$ on all of X^3 . It is defined on every ordered word, every global compatible-family coefficient, and every finite input translation. For this prescribed B_3 , and an added module included in the target with the displayed splitting, N_3 is minimal.

Proof. The zero-form unmerged output is a regular coefficient times $k\delta_{ik}\partial_{t_{ik}}$ with spectator J_j . The mixed zero-form outputs have vacuum in the pair block by (1.15). Thus all degree-2 outputs lie in M_3 .

Every v_g or q_g has vacuum states or quadratic current states, including their translations. Its normal coefficients are regular in the outer variable. The outer collision of a vacuum term is zero. In the Wick expansion of a quadratic current state against one current, a singular term uses exactly one contraction. It leaves exactly one translated current. Therefore every degree-1 output lies in S_3 . The same conclusion holds after all right derivatives.

The identity $Q\beta_3 + \beta_3 b = 0$ follows by expanding the defect. Since $QN_3 = 0$, it gives $\beta_3 b = 0$. Thus (1.16) is a complex, and direct substitution gives $QF_3 = F_3 \tilde{b}$.

For any pair choose a mixed word containing it and the coefficient q_{ij} . The corresponding defect is, up to its displayed sign, $k[\mathbf{1}\delta_{ij}|J_l]$. Its right differential closure is M_{ij} . It remains to show that S_3 is forced. Take $t = z_1 - z_2$, $s = z_2 - z_3$, $y = z_3$, and the global closed coefficient

$$\eta_{12} = \frac{dq_{12}}{t^2}. \quad (1.17)$$

On a face with constant relative order, its numerator is zero. On every other face the coefficient ring inverts t . Thus this is an actual member of C_3^1 . For $m = m_{12|3}$, its left edge coefficient is $g = 1/t^2$, and

$$q_g = \frac{k}{2}\mathbf{1}\delta_{12}\partial_t^2 + J^2\delta_{12}, \quad \beta_3(\eta_{12}m) = -2k(J\delta_{123}\partial_s + (TJ)\delta_{123}). \quad (1.18)$$

For the second formula, the inner vacuum term has zero outer output and $\text{pp}_s Y(J^2, s)J = 2k(J/s^2 + TJ/s)$. The orientation is $dz_1 \wedge dz_2 \wedge dz_3 = dt \wedge ds \wedge dy$. Multiplication by s gives $-2kJ\delta_{123}$. This nonzero element generates S_3 under right differential operators. Every added target submodule that absorbs the defect must therefore contain N_3 . $\square$

The factor t in q_g is decisive. The coefficient dq_{12}/t gives only $k\mathbf{1}\delta_{12}\partial_t$ and has zero outer collision. The first nonzero mixed outer output is (1.17), which has one additional inner pole. The new small-diagonal source term is exactly (1.18).

1.7 COHOMOLOGY RELATIVE TO THE ORIGINAL ORDERED COMPLEX

All cohomology in this section is cohomology of algebraic module complexes. Write $H_D^r = H^r(O_3) \otimes_{R_3} D_3$. The added module gives the short exact sequence

$$0 \longrightarrow N_3 \longrightarrow \tilde{P}_3 \longrightarrow P_3 \longrightarrow 0.$$

Its connecting maps are the induced defect maps $\delta_{-3} : H_D^{-3} \rightarrow M_3$ and $\delta_{-2} : H_D^{-2} \rightarrow S_3$. They are defined explicitly by (1.15) and right differential linearity.

For each $\sigma = ijk$, put $p_\sigma = t_{ij}t_{jk}$ and

$$z_\sigma = p_\sigma e_\sigma + t_{jk}q_{ij}m_\sigma + t_{ij}q_{jk}n_\sigma.$$

These are cycles, and

$$H^{-3}(O_3) = \bigoplus_{\sigma} (R_3 z_\sigma \oplus R_3 m_\sigma \oplus R_3 n_\sigma). \quad (1.19)$$

To prove completeness, a closed unmerged coefficient is constant on the full simplex. Its vertex values force it to belong to R_3 . For a cycle $f e_\sigma + g m_\sigma + h n_\sigma$, integration on its two actual edges gives $t_{ij} \mid f$

and $t_{jk} \mid f$. The two relatively prime linear polynomials give $p_\sigma \mid f$. Subtracting the corresponding multiple of z_σ leaves constant mixed coefficients. There are no incoming boundaries in degree -3 . This proves (1.19).

In fixed label order set

$$c = k[\mathbf{1}\delta_{12}\delta_{12}|J_3] - k[\mathbf{1}\delta_{13}\delta_{13}|J_2] + k[\mathbf{1}\delta_{23}\delta_{23}|J_1] \in M_3.$$

Every coordinate permutation here uses (1.9) and its density identification. If ε_σ is the sign of the permutation σ , then

$$\delta_{-3}(z_\sigma) = \varepsilon_\sigma p_\sigma c, \quad \delta_{-3}(m_\sigma) = \delta_{-3}(n_\sigma) = 0.$$

Indeed, $B_3 z_\sigma = p_\sigma [J_i | J_j | J_k]$ and $b z_\sigma = 0$. Apply Q and use its regular-function linearity.

This also gives a finite right-module presentation for the cohomology contribution from the pair diagonals. Put $t = z_1 - z_3$ and $s = z_2 - z_3$ in the next formula only. The six polynomials p_σ generate (t^2, ts, s^2) in R_3 . The relations $p_\sigma c$ therefore generate precisely

$$K = c t^2 D_3 + c t s D_3 + c s^2 D_3 \subset M_3. \quad (1.20)$$

The right differential closure removes this quotient completely.

LEMMA 1.7.1. One has $K = M_3$.

Proof. Let $g_{12} = [\mathbf{1}\delta_{12}|J_3]$ and temporarily put $x = t_{12}, s = t_{23}$. The element $c t_{13}^2 t_{23}^2$ belongs to K by (1.20). Its 13 and 23 components vanish, since a double-pole class is killed by its squared normal coordinate. On the 12 component, $t_{13} = x + s$ and normal multiplication gives

$$(c t_{13}^2 t_{23}^2)x = k g_{12} s^4.$$

Thus the class $\bar{g}_{12}$ in M_3/K satisfies $\bar{g}_{12} s^4 = 0$. It also satisfies $\bar{g}_{12} P = 0$, where $P = \partial_1 + \partial_2$. The Weyl relation is $[P, s] = 1$. If $\bar{g}_{12} s^r = 0$ for $r > 0$, then

$$0 = (\bar{g}_{12} s^r) P = (\bar{g}_{12} P) s^r - r \bar{g}_{12} s^{r-1} = -r \bar{g}_{12} s^{r-1}.$$

Descending from $r = 4$ gives $\bar{g}_{12} = 0$. The two relabelings give the same conclusion for g_{13} and g_{23} . These three elements generate M_3 , proving the assertion. $\square$

THEOREM 1.7.2. Both connecting maps $\delta_{-3} : H_D^{-3} \rightarrow M_3$ and $\delta_{-2} : H_D^{-2} \rightarrow S_3$ are surjective. The entire cohomology change from P_3 to $\tilde{P}_3$ is described by

$$\begin{aligned} 0 &\longrightarrow H^{-3}(\tilde{P}_3) \longrightarrow H_D^{-3} \xrightarrow{\delta_{-3}} M_3 \longrightarrow 0, \\ 0 &\longrightarrow H^{-2}(\tilde{P}_3) \longrightarrow H_D^{-2} \xrightarrow{\delta_{-2}} S_3 \longrightarrow 0, \\ H^r(\tilde{P}_3) &\cong H_D^r \quad (r \neq -3, -2). \end{aligned} \quad (1.21)$$

The first kernel has the explicit equation $\sum_\sigma \varepsilon_\sigma p_\sigma c A_\sigma = 0$ for the coefficients $A_\sigma \in D_3$ of z_σ . Every mixed basis coefficient is unrestricted there.

Proof. The element $\eta_{12}m_{12|3}$ is closed because $d_C\eta_{12} = 0$ and $bm_{12|3} = 0$. Its connecting image is (1.18). Multiplication by s gives a nonzero scalar times the cyclic generator of S_3 . Right differential linearity proves surjectivity. In particular, this closed element is not a boundary in P_3 . Otherwise its defect would vanish, since $\beta_3b = 0$.

The long exact sequence has the consecutive terms

$$0 \rightarrow H^{-3}(\tilde{P}_3) \rightarrow H_D^{-3} \rightarrow M_3 \rightarrow H^{-2}(\tilde{P}_3) \rightarrow H_D^{-2} \rightarrow S_3 \rightarrow H^{-1}(\tilde{P}_3) \rightarrow H_D^{-1} \rightarrow 0.$$

Lemma 1.7.1 gives surjectivity of δ_{-3} . Use both surjections in this sequence. All other degrees follow because N_3 has only the two stated components. $\square$

These formulas separate two changes. The translation construction replaces $H^r(O_3)$ by the exact tensor product in (1.6). The native collision terms then impose (1.21). The vanishing $M_3/K = 0$ uses the vacuum tangent relation and Weyl commutators. A quotient taken only over regular functions does not permit this argument. The coefficient integral table specifies δ_{-2} on every cycle, without requiring a coefficient-formality assertion about C_3 . Both connecting maps are nonzero, so the projection is not a quasi-isomorphism.

1.8 THE FIRST FOUR-CURRENT COMPATIBILITY

The binary added module has vacuum output. Tensoring it with a single current gives the ternary pair module through the corresponding closed direct image. Its differential remains zero by the vacuum calculation above. Tensoring with two currents introduces a new collision.

In E_4 , for $i < j$ and the remaining labels $l < m$, set

$$w_{ij} = k[\mathbf{1}\delta_{ij}\partial_{t_{ij}}|J_l|J_m].$$

For a decomposition into two pairs, order the pair blocks lexicographically and let

$$d_{ij|lm} = k^2[\mathbf{1}\delta_{ij}\partial_{t_{ij}}|\mathbf{1}\delta_{lm}\partial_{t_{lm}}].$$

These have degrees -3 and -2 , respectively. They are supported on their actual pair and double-pair diagonals. The double-pair class is nonzero because both transfer derivatives and k^2 are nonzero. There is no outer pole between the two vacuum blocks.

PROPOSITION 1.8.1. The six pair terms cannot all be assigned zero differential while retaining their native images. The smallest right differential subcomplex containing them includes the three double-pair modules. Its differentials on the displayed generators are

w	w_{12}	w_{34}	w_{13}	w_{24}	w_{14}	w_{23}	(1.22)
Qw	$-d_{12 34}$	$d_{12 34}$	$-d_{13 24}$	$d_{13 24}$	$-d_{14 23}$	$d_{14 23}$	

Every double-pair term is closed. The resulting correction satisfies the four-current equation $Q^2[J_1|J_2|J_3|J_4] = 0$.

Proof. In $[\mathbf{1}_{ij}|J_l|J_m]$, both collisions involving the vacuum have zero principal part. The remaining term in (1.9) is $[\mu(J_l, J_m)|\mathbf{1}_{ij}]$ with positive sign. Putting its blocks in the prescribed order gives precisely (1.22). Right derivatives preserve these equations. Multiplying the two normal coordinates lowers the two derivatives to the simple double-delta generator. Its right differential closure is the full regular

double-pair vacuum module. Thus these three modules are forced, and they suffice to close the displayed current subspace under Q . Their next differential vanishes because both states are vacuum with regular coefficients.

For four even inputs, the first collision of labels $i < j$ has sign $(-1)^{i+j-3}$ when its block is placed first. Hence

$$Q[J_1|J_2|J_3|J_4] = w_{12} - w_{13} + w_{14} + w_{23} - w_{24} + w_{34}.$$

Each complementary pair has the same first sign and opposite second signs in (1.22). The three double-pair coefficients cancel individually. $\square$

This calculation specifies the next differential on the regular four-current collision subspace. The binary and ternary maps above have their full stated global coefficient domains. A map on every four-input ordered coefficient requires additional operations on the other four-point partition summands. An all-arity bar coalgebra morphism also requires ordered-surjection and deconcatenation identities. Those identities are not consequences of the individual cochain maps constructed here.

BIBLIOGRAPHY

- [1] A. Beilinson and V. Drinfeld, *Chiral Algebras*, American Mathematical Society, 2004, §3.1.1–3.1.2 and §3.4.11, equation (3.4.11.1), pp. 157–158, 182–183.